

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

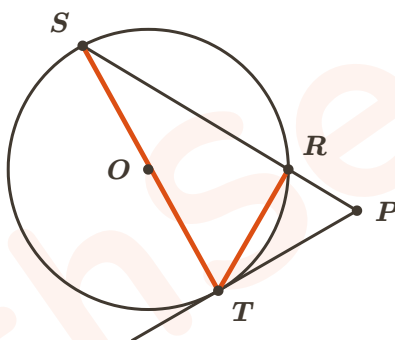
Chapter 18: Tangents and Intersecting Chords

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Tangent and secant from an external point

1. PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**



SOLUTION

(a) In $\triangle PTR$ and $\triangle PST$: $\angle P$ is common, and $\angle PTR = \angle PST$ (angle between tangent and chord equals the angle in the alternate segment). So $\triangle PTR \sim \triangle PST$ (AA).

(b) Corresponding sides are proportional: $\frac{PT}{PS} = \frac{PR}{PT}$, so $PT^2 = PR \times PS$.

(c) $PT^2 = 4 \times 16 = 64$

Answer: $PT = 8$ cm

2. A circle has centre O and radius 9 cm. The diameter AB is produced to a point P , and PQ is a tangent from P . If $PA = 24$ cm, find PQ . **ICSE 2024, 3 marks**

SOLUTION

$AB = 18$ cm, so $PB = PA - AB = 24 - 18 = 6$ cm.

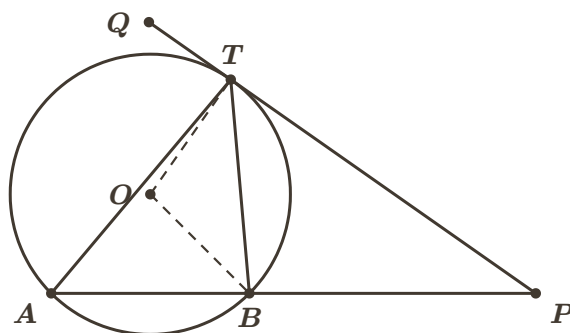
$PQ^2 = PA \times PB = 24 \times 6 = 144$

Answer: $PQ = 12$ cm

(Check: $OP = 15$ cm and $OQ = 9$ cm, so $PQ = \sqrt{15^2 - 9^2} = 12$ cm.)

3. PQ is a tangent to a circle with centre O at T . The chord AB produced meets the tangent at P . $AB = 9$ cm, $BP = 16$ cm, $\angle PTB = 50^\circ$ and $\angle OBA = 45^\circ$. Find (a) PT (b) $\angle BAT$ (c) $\angle BOT$ (d) $\angle ABT$.

ICSE 2023, 4 marks



SOLUTION

(a) $PA = 9 + 16 = 25$ cm. $PT^2 = PB \times PA = 16 \times 25 = 400$, so $PT = 20$ cm.

(b) By the alternate segment theorem, $\angle BAT = \angle PTB = 50^\circ$.

(c) Angle at the centre is twice the angle at the circumference: $\angle BOT = 2 \times 50^\circ = 100^\circ$.

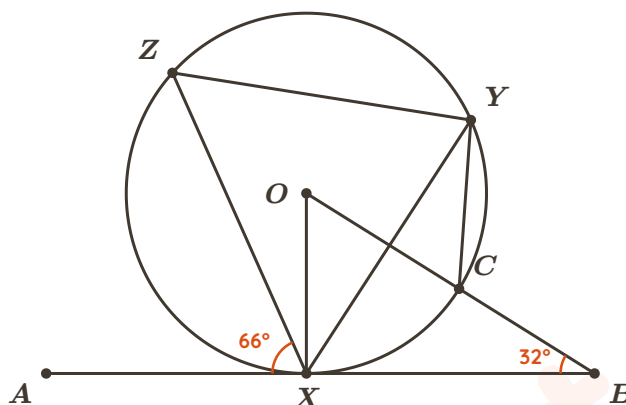
(d) $OB = OT$, so $\angle OBT = \frac{180^\circ - 100^\circ}{2} = 40^\circ$. $\angle ABT = \angle OBA + \angle OBT = 45^\circ + 40^\circ = 85^\circ$.

Answer: (a) 20 cm (b) 50° (c) 100° (d) 85°

2 The alternate segment theorem

4. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$.

ICSE 2025, 4 marks



SOLUTION

(a) $OX \perp AB$ (radius and tangent). In $\triangle OXB$: $\angle BOX = 90^\circ - 32^\circ = 58^\circ$.

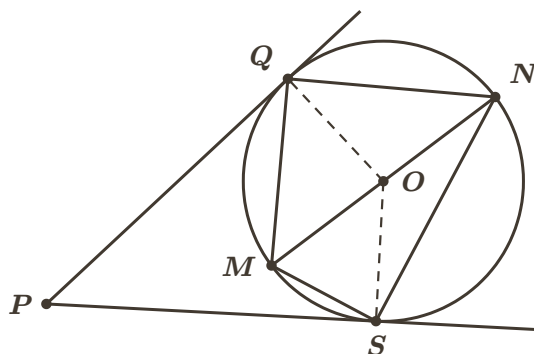
(b) The arc CX subtends $\angle COX = 58^\circ$ at the centre and $\angle CYX$ on the circle: $\angle CYX = \frac{58^\circ}{2} = 29^\circ$.

(c) By the alternate segment theorem, $\angle ZYX = \angle AXZ = 66^\circ$.

(d) $ZY = XY$, so $\angle YXZ = \frac{180^\circ - 66^\circ}{2} = 57^\circ$. Also $\angle OXZ = 90^\circ - 66^\circ = 24^\circ$. So $\angle OXY = 57^\circ - 24^\circ = 33^\circ$.

Answer: (a) 58° (b) 29° (c) 66° (d) 33°

5. PR and PT are tangents from P to a circle with centre O , touching it at Q and S . MN is a diameter, with M on the arc QS nearer to P . Given $\angle PQM = 42^\circ$ and $\angle PSM = 25^\circ$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. **ICSE 2024, 4 marks**



SOLUTION

(a) $OQ \perp PR$ (radius and tangent), so $\angle OQM = 90^\circ - 42^\circ = 48^\circ$.

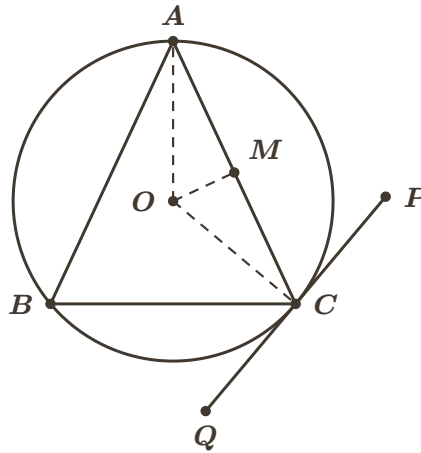
(b) By the alternate segment theorem, $\angle QNM = \angle PQM = 42^\circ$ and $\angle SNM = \angle PSM = 25^\circ$. So $\angle QNS = 42^\circ + 25^\circ = 67^\circ$.

(c) Angle at the centre is twice the angle at the circumference: $\angle QOS = 2 \times 67^\circ = 134^\circ$.

(d) $QNSM$ is a cyclic quadrilateral: $\angle QMS = 180^\circ - 67^\circ = 113^\circ$.

Answer: (a) 48° (b) 67° (c) 134° (d) 113°

6. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O . PQ is the tangent at C , with Q on the side of BC away from A . OM is perpendicular to the chord AC and $\angle COM = 65^\circ$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. **ICSE 2024, 3 marks**



SOLUTION

- (a) The perpendicular from the centre bisects the chord and the angle at the centre: $\angle AOC = 2 \times 65^\circ = 130^\circ$. So $\angle ABC = \frac{130^\circ}{2} = 65^\circ$.
 (b) $AB = AC$, so $\angle ACB = \angle ABC = 65^\circ$ and $\angle BAC = 180^\circ - 130^\circ = 50^\circ$.
 (c) By the alternate segment theorem, $\angle BCQ = \angle BAC = 50^\circ$.

Answer: (a) 65° (b) 50° (c) 50°

7. RT is a tangent to a circle at S . P and Q are points on the circle with $\angle PST = 30^\circ$ and $\angle SPQ = 60^\circ$. Find $\angle PSQ$. (a) 40° (b) 30° (c) 60° (d) 90° **ICSE 2023, 1 mark**

SOLUTION

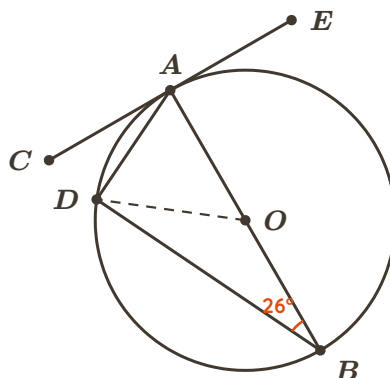
By the alternate segment theorem, $\angle PQS = \angle PST = 30^\circ$.

In $\triangle PQS$: $\angle PSQ = 180^\circ - 60^\circ - 30^\circ = 90^\circ$

Answer: (d)

8. AB is a diameter of a circle with centre O , and CE is the tangent at A . D is a point on the circle on the same side of AB as C , with $\angle ABD = 26^\circ$. Find (a) $\angle BDA$ (b) $\angle BAD$ (c) $\angle CAD$ (d) $\angle ODB$.

ICSE 2023, 4 marks



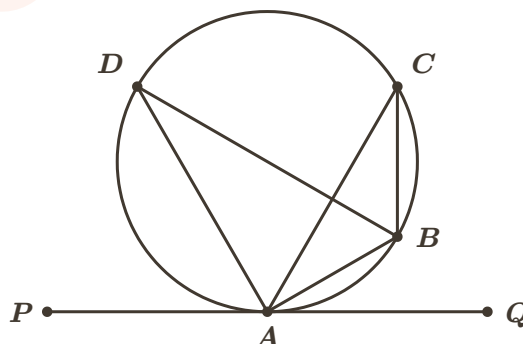
SOLUTION

- (a) $\angle BDA = 90^\circ$ (angle in a semicircle).
 (b) $\angle BAD = 180^\circ - 90^\circ - 26^\circ = 64^\circ$
 (c) By the alternate segment theorem, $\angle CAD = \angle ABD = 26^\circ$.
 (d) $OD = OB$ (radii), so $\angle ODB = \angle OBD = 26^\circ$.

Answer: (a) 90° (b) 64° (c) 26° (d) 26°

9. PQ is a tangent to a circle at A . B, C and D are points on the circle such that AB bisects $\angle CAQ$ and AD bisects $\angle PAC$. If $\angle BAQ = 30^\circ$, prove that (a) BD is a diameter (b) $\triangle ABC$ is isosceles.

ICSE 2017, 3 marks



SOLUTION

- (a) $\angle CAQ = 2 \times 30^\circ = 60^\circ$, so $\angle BAC = 30^\circ$ and $\angle PAC = 180^\circ - 60^\circ = 120^\circ$.
 AD bisects $\angle PAC$, so $\angle CAD = 60^\circ$. Then $\angle BAD = \angle BAC + \angle CAD = 30^\circ + 60^\circ = 90^\circ$.
 A chord that subtends a right angle at the circumference is a diameter, so BD is a diameter.
 (b) By the alternate segment theorem, $\angle ACB = \angle BAQ = 30^\circ$. Also $\angle BAC = 30^\circ$.
 Two angles of $\triangle ABC$ are equal, so $AB = BC$. **Both results are proved.**

3 Tangent properties

10. PS and PT are tangents to a circle from P , with $\angle SPT = 80^\circ$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) 140° (b) 90° (c) 80° (d) 50° **ICSE 2024, 1 mark**

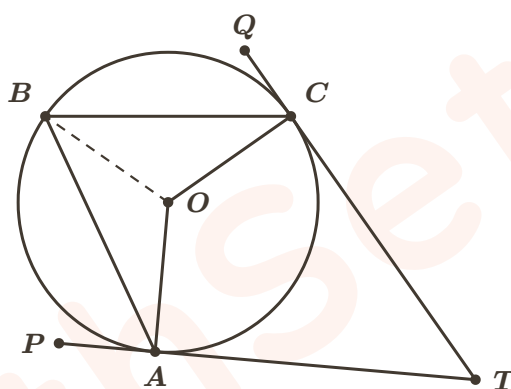
SOLUTION

$$PS = PT, \text{ so } \angle PTS = \angle PST = \frac{180^\circ - 80^\circ}{2} = 50^\circ.$$

$SQ \parallel PT$ and ST is a transversal, so $\angle QST = \angle PTS = 50^\circ$ (alternate angles).

Answer: (d)

11. TP and TQ are tangents to a circle with centre O , touching it at A and C . B is a point on the circle with $\angle BCQ = 55^\circ$ and $\angle BAP = 60^\circ$. Find (a) $\angle OBA$ and $\angle OBC$ (b) $\angle AOC$ (c) $\angle ATC$. **ICSE 2020, 4 marks**



SOLUTION

(a) $OA \perp TP$, so $\angle OAB = 90^\circ - 60^\circ = 30^\circ$. $OA = OB$, so $\angle OBA = 30^\circ$.

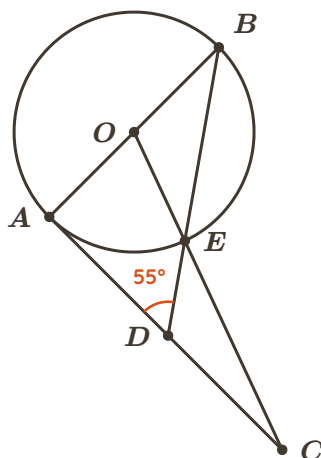
$OC \perp TQ$, so $\angle OCB = 90^\circ - 55^\circ = 35^\circ$. $OB = OC$, so $\angle OBC = 35^\circ$.

(b) $\angle ABC = 30^\circ + 35^\circ = 65^\circ$, so $\angle AOC = 2 \times 65^\circ = 130^\circ$.

(c) In quadrilateral $OATC$ the angles at A and C are 90° , so $\angle ATC = 180^\circ - 130^\circ = 50^\circ$.

Answer: (a) 30° and 35° (b) 130° (c) 50°

12. AB is a diameter of a circle with centre O , and AC is the tangent at A . D is a point on AC , and BD cuts the circle at E . The line OE produced meets the tangent at C . If $\angle ADB = 55^\circ$, find $x = \angle ACO$ and $y = \angle AOE$. **ICSE 2019, 3 marks**



SOLUTION

$\angle BAD = 90^\circ$ (the tangent is perpendicular to the radius at the point of contact).

In $\triangle ABD$: $\angle ABD = 180^\circ - 90^\circ - 55^\circ = 35^\circ$.

$y = \angle AOE = 2 \times \angle ABE = 70^\circ$ (the angle at the centre is twice the angle at the circumference).

In $\triangle OAC$: $x = 180^\circ - 90^\circ - 70^\circ = 20^\circ$.

Answer: $x = 20^\circ, y = 70^\circ$

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